A Comparative Study of Users' Information-Seeking Practices Across Search Engines and Generative AI Chatbots

Elsa Lichtenegger
University of Zurich
Zurich, Switzerland
lichtenegger@ifi.uzh.ch

Aleksandra Urman
University of Zurich
Zurich, Switzerland
urman@ifi.uzh.ch

Aniko Hannak
University of Zurich
Zurich, Switzerland
hannak@ifi.uzh.ch

Abstract

Web search engines have long been the primary gateway to online information, but generative AI chatbots are emerging as alternative tools. Yet we lack empirical understanding of how users choose between these tools and how their information-seeking behaviors differ across these contexts. Understanding how users select, use, and assess information retrieval (IR) systems is essential for evaluating them in ways that reflect user expectations across contexts. To shed light on user perceptions and usage of IR systems, we surveyed 84 participants about their recent search engine and GenAI chatbot use, analyzing task types, contextual triggers, outcomes, and tool preferences. Our findings reveal that users approach search engines and GenAI chatbots with distinct philosophies. With search engines, they follow a 'Browse and Verify' approach, prioritizing reliability and source verifiability. With GenAI chatbots, they adopt a 'Delegate and Consume' approach, valuing quick summarization and personalized content. Despite these stated preferences, actual usage shows that users frequently turn to GenAI chatbots for actionable guidance in high-stakes domains such as health and relationship matters, hinting at a divergence between stated philosophies and real-world behavior. These findings underscore the need for IR systems that support reliable information assessment, transparent sources, and user-guided evaluation.

CCS Concepts

• Information systems → Users and interactive retrieval; **Web search engines**; **• Human-centered computing → Empirical studies in HCI**.

Keywords

Web search; Search engines; Generative AI; Information-seeking; User behavior

ACM Reference Format:
Elsa Lichtenegger, Aleksandra Urman, and Aniko Hannak. 2026. A Comparative Study of Users' Information-Seeking Practices Across Search Engines and Generative AI Chatbots. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '26), July 20–24, 2026, Melbourne, VIC, Australia.* ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3805712.3809525

SIGIR '26, Melbourne, VIC, Australia

ACM ISBN 979-8-4007-2599-9/2026/07
https://doi.org/10.1145/3805712.3809525

1 Introduction

Web search engines (SEs) have long served as the primary interface for online information retrieval (IR). Early systems relied on search engine result pages (SERPs) composed of ranked lists of links [9, 10]. Over time, SEs increasingly incorporated SERP features such as direct answers and knowledge panels, thereby providing faster access to information [38]. Recently, GenAI chatbots[1] have emerged as alternatives to traditional search. These systems offer synthesized answers, support conversational interaction, and allow users to iteratively refine their information needs through follow-up prompts [39, 28]. Their rapid adoption for information-seeking tasks has prompted major SE providers to integrate generative components into their systems, including AI-powered summaries and conversational search modes. At the same time, GenAI chatbots are incorporating retrieval capabilities, such as access to web content and source linking. As a result, retrieval-based and generative systems are increasingly converging, and the boundary between them is becoming less distinct.This convergence is reflected, for instance, in how public discourse increasingly frames GenAI chatbots as potential successors to traditional SEs or even as some sort of *"super search engines"* [28, 57, 33, 6]. Empirical evidence points in the same direction: a substantial share of GenAI chatbot queries involve information seeking or practical guidance, tasks historically associated with search engines [8].

This system-level convergence raises an important question on the user side: even as these tools are used for similar information-seeking tasks, do users perceive and approach them as interchangeable, or do distinct mental models and interaction strategies persist despite their growing functional overlap? Understanding how users conceptualize and use these systems is particularly important, as they were originally designed for different purposes. SEs are built to retrieve and rank relevant information from external sources, supporting comparison and verification, whereas LLM-based systems are designed to generate fluent, contextually appropriate responses based on learned patterns in data. As a result, LLMs can produce plausible but not necessarily accurate content [1, 2, 29, 32, 63]. Prior work has highlighted risks such as hallucinations, bias, and overreliance [64, 69], which are further amplified by users' tendency to trust fluent and authoritative-sounding responses [21, 34].

While these challenges highlight the importance of understanding how users engage with these tools, most existing research has focused on system-level properties, such as answer quality, retrieval augmentation, and citation accuracy, or on technical risks [15, 23, 68]. As a result, we lack a clear understanding of how users navigate between SEs and GenAI chatbots in real-world information-seeking

[1]In this paper, 'GenAI chatbots', 'AI', etc., refer specifically to generative LLM-based tools, primarily chatbots such as ChatGPT.

contexts, and whether convergence at the system level translates into convergence at the user level. Studying users' mental models and interaction strategies across both paradigms informs how users choose tools, form expectations, and navigate risks, guiding the design and evaluation of IR systems that align with user needs and support informed use.

In this paper, we address this gap through a comparative analysis of how users choose between SEs and GenAI chatbots and how their information-seeking behaviors differ across these tools. Specifically, we address the following research questions:

RQ1: What patterns emerge in how users choose between SEs and GenAI chatbots?
RQ2: How do information-seeking sessions and task contexts differ across these tools?
RQ3: How do users' stated expectations and preferences align with their actual usage patterns?

To answer these questions, we conducted a survey with 84 participants in which they retrieved and reflected on their recent information-seeking sessions with SEs and GenAI chatbots, resulting in 500 reported sessions. This approach allows for insights into user motivations, expectations, task approaches, and tool preferences across real-world information-seeking contexts.

With this work, we provide three contributions to IR research. **(1)** We establish a baseline understanding of how users perceive and use retrieval-based versus generative IR tools. Despite growing functional overlap between SEs and GenAI chatbots, we find that users continue to rely on two distinct approaches when selecting which tool to use: a 'Browse and Verify' philosophy for SEs emphasizing reliability and source verification, and a 'Delegate and Consume' philosophy for GenAI chatbots emphasizing quick, ready-made answers and personalization. We further document specific information-seeking task types associated with each tool. **(2)** We uncover a potential misalignment between users' stated preferences and their observed usage patterns. While users associate SEs with source verification and reliability and GenAI chatbots with efficiency, they nevertheless frequently turn to GenAI chatbots for action-oriented advice in sensitive domains such as health and relationship topics, hinting at a divergence between conceptualized and actual tool use. **(3)** We derive concrete implications for the evaluation and benchmarking of IR systems from our findings. By studying users at a critical inflection point in February 2025, before AI-powered features were fully integrated into SEs, we provide insights into how effective, reliable, and critically engaged IR use is being reshaped as generative capabilities become increasingly embedded in everyday search experiences.

2 Related Work

SEs have long served as primary gateways to online information [7]. Dominant platforms like Google rank among the most visited websites [49, 47] and control substantial search market share, reinforced by default settings on browsers and devices [13, 31, 19]. This prominence makes SEs not only influential for information access but also for shaping user attention and perception through the ranking and presentation of results, increasingly satisfying information needs directly on the SERP via knowledge panels and featured snippets [38, 17, 14]. Users tend to trust highly ranked results and often rely on them without further verification [16, 40], a behavior that can shape topic understanding, real-world decisions, and opinions [37, 12]. At the same time, algorithmic curation and personalization can reinforce biases, limit exposure to diverse perspectives, and reduce transparency, highlighting the gatekeeping role of SEs on information access [62, 36, 3].

The public release of ChatGPT in late 2022 reshaped the IR landscape by making LLM-based conversational systems widely accessible [20]. While not designed as SEs, their capacity to answer questions, summarize content, and generate text has made them increasingly used for tasks traditionally served by web search [39, 46, 8]. The rise of these systems has prompted SEs to integrate generative AI features, such as Google's AI Overviews and AI Mode, further blurring the line between retrieval and conversation [44, 55, 51]. The widespread adoption and integration of these systems have shifted the IR landscape, raising questions about user behavior and interaction in emerging information-seeking paradigms.

Emerging evidence indicates that LLM-powered conversational interfaces change how people approach information-seeking. Schneider [45] found that user prompts initially resemble task-oriented commands but then become more conversational and polite. Schneider et al. [46] showed that participants refine their goals over multiple conversational turns, with the conversational format supporting exploration of unfamiliar topics. This reflects a cognitive shift: users perceive LLM systems as collaborative partners rather than mere tools. Comparative work further suggests that users engage with conversational LLMs differently from traditional web search. Suri et al. [56] found that users employ generative search for cognitively complex tasks across professional, scientific, technical, and educational domains. However, research highlights important trade-offs. Users often over-rely on AI-generated responses, reducing fact-checking and critical evaluation [52, 67]. Work on explainability further shows that users struggle to detect inaccuracies in conversational responses without explicit interface support [25].

While prior work has shown that LLM-based conversational search alters information-seeking behavior and has begun to compare usage across these systems and SEs, existing studies have largely relied on lab settings [67] or interaction logs [56]. These approaches capture behavioral traces but not the mental models, motivations, and perceived differences that shape how users approach each system in real-world contexts. Our comparative study addresses this gap by surveying users about their real-world information-seeking sessions across SEs and GenAI chatbots, capturing not only what they did but how they conceptualized and motivated their tool choices.

3 Methods

To study how users interact with GenAI chatbots and SEs in everyday contexts, we designed a survey in which participants reviewed their last three interactions with each platform and reflected on them, capturing both actual usage and their perceptions [27].

3.1 Survey Design

The survey consists of three main parts: (1) Questions on general SE and GenAI chatbot usage patterns. (2) A guided reflection exercise on users' recent interactions with both platforms. (3) Questions on digital literacy [18], AI knowledge [35], attitudes towards AI [50] and SEs [60], political attitudes [24], and demographics.

To ensure consistent understanding, we define key terms throughout the survey. *GenAI chatbots* are defined as computer programs using artificial intelligence to engage in conversations on a wide range of topics (e.g., ChatGPT, Gemini, Microsoft Copilot), explicitly excluding personal assistants and customer service bots. A *SE session* is defined following Jansen et al. [22] as *'a series of interactions by the user toward addressing a single information need'* while an *AI session* denotes a single, continuous interaction with a GenAI chatbot focused on one task or topic.

3.2 Core Reflection Exercise

In the central component of our study, participants are asked to review their last three SE sessions and their last three AI sessions. We provide step-by-step instructions to retrieve past histories from SEs and GenAI chatbots. By having users review their recent histories, we aim to reduce recall bias while capturing routine, ecologically valid behaviors, rather than relying on atypical or selectively remembered episodes [5, 58].

To capture both observable behaviors and underlying cognitive processes, participants provide the initial query or prompt for each session and complete reflection questions based on an extended version of Xie [66]'s search goals framework, tailored to SEs and GenAI chatbots:

- *Current goal:* What specific information or output were you trying to obtain?
- *Leading goal:* What situation or context led you to initiate this session?
- *Long-term goal:* What was the underlying purpose or larger objective?
- *Interactive intention:* What steps or actions did you take during the session?
- *Outcome realization:* What did you do with the information or output?

Additionally, participants explain their tool choice by reflecting on whether the same goal could have been achieved using the other tool, followed by an open-ended justification. The wording of these questions is adapted depending on the session type, such that participants compare the focal tool (i.e., SE or GenAI chatbot) against the alternative.

- Do you think you could have used [alternative tool] instead of [focal tool] to achieve the goal of this session? (Y/N)

Depending on their response, participants provide an open-ended explanation:

- Why did you use [focal tool] instead of [alternative tool]? *or*
- Why do you think only [focal tool], and not [alternative tool], could achieve the goal of this task?

3.3 Participants and Procedure

In February 2025, we recruited 84 US-based participants through Prolific and administered the study via Qualtrics. Eligibility required prior experience with SEs and GenAI chatbots, as well as desktop use, such that participants could easily access their interaction histories. A screening step verified participants' ability to retrieve these histories using tool-specific instructions for major platforms (Google, Yahoo, Bing for SEs; ChatGPT, Gemini, Microsoft Copilot, Perplexity, Claude for GenAI chatbots). After verification, participants selected their most frequently used SE and AI chatbot for the study. Most used Google as their SE (n = 83; Bing: n = 1) and ChatGPT as their GenAI chatbot (n = 73; Gemini: n = 5; Claude: n = 3; Microsoft Copilot: n = 3). Google is the dominant SE (~87% US market share, [54]) and ChatGPT the most widely used GenAI chatbot (~88% US market share, [53]) at the time of the study. This aligns with our sample distribution, supporting the use of SE and GenAI chatbot as umbrella terms. Next, participants completed baseline usage questions, a guided reflection exercise, and skill-based as well as demographic questions. All responses underwent manual quality control. We excluded sessions that exhibited signs of low effort (e.g., incomplete or uninformative responses, such as expressions of uncertainty or lack of recall) or automated generation (e.g., unusually long, overly polished, or stylistically AI-generated responses). This resulted in 241 SE sessions and 247 AI sessions for analysis.

Demographics, Usage, and Attitudes. Participants average 37.48 years of age (SD = 12.98, median = 34.5); 54.8% identify as female and 45.2% as male. 59.8% hold at least a bachelor's degree, 23% report some college or an associate/technical degree, and 8.7% a high school diploma or less. Most participants use SEs daily (76.2%) with 1-20 searches per day (66.7%). GenAI chatbot use was more varied, ranging from daily (32.1%) to several times a week (29.8%), weekly (11.9%), occasional (20.2%), and rare use (6.0%).

Participants exhibit high digital literacy (5-point scale; M = 4.05, SD = 0.70, Cronbach's α = 0.894) [18] and neutral-to-slightly positive attitudes toward AI (7-point scale; M = 4.90, SD = 0.96, Cronbach's α = 0.666) [50]. Perceived independence of SEs from political or governmental influence is moderate (7-point scale; M = 3.66, SD = 1.67, inter-item correlation r = 0.704, $p < 0.001$) [60], as is knowledge of GenAI (6-item true/false; M = 2.90, SD = 1.72, Cronbach's α = 0.681) [35]. Political ideology, measured on an 11-point liberal-conservative scale, shows a slight liberal lean (M = 4.82, SD = 2.87) [24].

3.4 Analysis Approach

Preference Types. To assess tool preferences, participants indicated whether their session goal could have been achieved with the alternative tool (GenAI chatbot vs. SE) using a yes/no question, followed by an open-ended explanation. Participants' explanations often reflected subjective interpretations of tool capabilities, task fit, and effort rather than a strict assessment of feasibility. For example, writing an email can theoretically be achieved using either tool: a SE can provide relevant examples, templates, or phrasing guidance that users can adapt, while a GenAI chatbot can directly generate a full draft. However, some participants may perceive SEs as less suitable for such goals, as the interaction typically involves a multi-step process of searching, selecting, and synthesizing information rather than directly producing a complete output. Due to this interpretive ambiguity, we analyze all explanations together, regardless of the initial yes/no response.

First, we inductively developed a codebook for dataset annotation: we identified preference reason categories through iterative coding and discussion of a subset of the open-ended responses, adding new categories when recurring themes emerged. Multiple categories could be assigned per response, as participants often stated several distinct justifications. To assess inter-rater reliability (IRR), we defined agreement as both coders independently assigning identical sets of categories to a response (i.e., fully overlapping category sets). Reliability was calculated on a subset of AI sessions (N = 80, IRR = 0.90) and SE sessions (N = 56, IRR = 0.89). After reaching a satisfactory level of agreement, two coders annotated the full sample.

Preference Analysis. Following the development of preference categories, we compared preference distributions between AI and SE sessions, descriptively examining category frequencies and conceptual differences. To examine how preference reasons clustered, we conducted co-occurrence network analyses on multi-coded responses using R (version 4.4.0) [43], with packages igraph [11], tidygraph [42], and ggraph [41]. Co-occurrence matrices captured pairs of reasons appearing within the same response, retaining only pairs with frequency $\geq$ 2 to focus on meaningful patterns while reducing noise. Nodes represented preference reasons and edges their co-occurrence. The SE preference network comprised 10 nodes and 14 edges, while the AI preference network contained 11 nodes and 21 edges. We identified thematic clusters using Louvain community detection [4], selected for its suitability in exploratory analyses without predefined cluster numbers. To assess robustness, we applied the Leiden algorithm [59], yielding identical community structures (SE: 3 communities, modularity = 0.304; AI: 4 communities, modularity = 0.159). Given these small-to-moderate modularity scores, we also examined cross-cluster connections (frequency $\geq$ 2) and qualitatively compared network structure, cluster composition, and central nodes across tools.

Trigger, Topic, Task, and Outcome Types. To capture how participants used GenAI chatbots and SEs in practice, we analyzed full session reflections, including the initial prompt or query, stated goals, and downstream actions. Unlike prior work that focuses on prompts alone, this session-level perspective reveals broader intentions, such as exploratory learning or problem solving, that are often not evident from the initial query. We relied strictly on participants' written reflections without inferring unstated actions. From this holistic analysis, we developed a four-dimensional coding scheme: *(1) Topic*: the subject domain; *(2) Task*: the underlying task intent; *(3) Outcome*: the downstream action after information retrieval; *(4) Trigger*: the situation initiating the session.

Task coding began with established categories, namely lookup and exploratory [30, 65], and guidance [26], and was extended with a generative category (producing new content). Topic, outcome, and trigger were coded inductively, with categories refined through iterative discussion after pilot coding 20 sessions. The finalized coding scheme was independently applied by two researchers to a subset of sessions (47 AI, 46 SE). Inter-rater reliability was assessed using Cohen's kappa. Agreement was high across all dimensions (AI: topic, Cohen's κ = 0.93, CI: 0.85-1, task, κ = 0.95, CI: 0.86-1, trigger, κ = 0.94, CI: 0.87-1, outcome, κ = 0.89, CI: 0.79-0.99; SE: topic, κ = 0.90, CI: 0.81-0.99, task κ = 0.83, trigger κ = 0.90, CI: 0.78-1, CI: 0.69-0.97, outcome κ = 0.96, CI: 0.89-1). The finalized coding scheme was then applied to all remaining sessions and is reported in the results section. Differences in category distributions

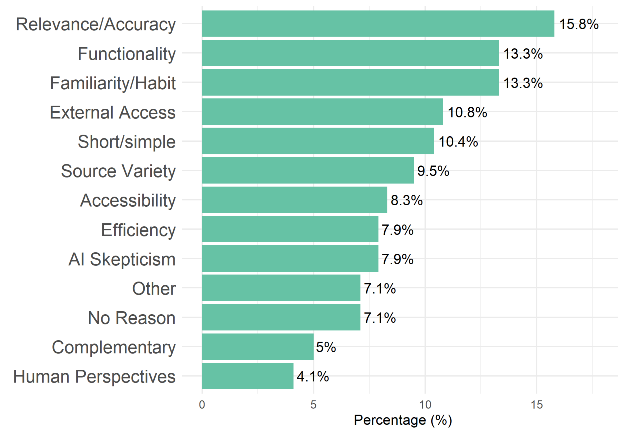

Figure 1: Preferences for using SEs

between tools were tested using χ^2 tests of independence for each dimension. For significant results, standardized residuals were used to identify driving categories (significant if |residual| > 2).

Session Flow Analysis. We examined within-session associations between triggers, tasks, and outcomes, excluding topics due to sparse cells resulting from numerous categories. For each relationship and modality, we calculated conditional probabilities ($P(task \mid trigger)$, $P(outcome \mid task)$, $P(outcome \mid trigger)$). Overall associations were assessed using χ^2 tests with Cramér's V as a measure of effect size. Given that several analyses violated χ^2 assumptions (>20% of cells with expected frequencies <5), post-hoc pairwise comparisons used Fisher's exact tests with Benjamini–Hochberg correction (α = 0.05) to control the false discovery rate.

4 Results

4.1 Tool Choice

4.1.1 Preference Categories. Our analysis of participants' stated preferences for SEs reveals several distinct categories, with relative frequencies across 241 sessions shown in Figure 1. Participants most frequently cite *Relevance/Accuracy* (15.8%), perceiving SE results as closely aligned with their information needs and verifiable through traceable sources. *Functionality* (13.3%) is another key factor, reflecting tasks seen as incompatible with GenAI chatbots, such as searching for images or videos, using specific features like currency converters, or completing activities such as shopping. *Familiarity/Habit* (13.3%) also guides choice, with participants often defaulting to SEs without consciously evaluating alternatives. *External Access* (10.8%) reflects using SEs to reach specific websites, while *Short/Simple* (10.4%) describes when the task at hand is perceived as straightforward. *Source Variety* (9.5%) corresponds to the ability to access and compare multiple sources independently, and *Accessibility* (8.3%) is cited when SEs are perceived as easy to access in the moment, already open in the workflow, or set as a technical default, e.g., in a browser. *AI Skepticism* (7.9%) reflects users' mistrust of GenAI chatbots due to prior negative experiences or concerns about functionality, while *Efficiency* (7.9%) highlights the desire for quick,

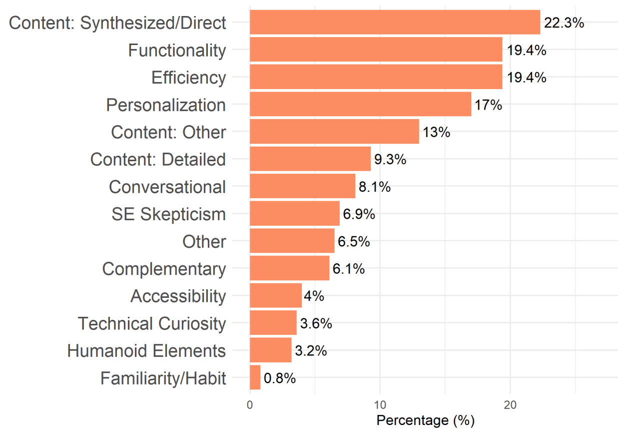

Figure 2: Preferences for using GenAI chatbots

direct results. Other vague or unusual reasons (*Other; 7.1%*) include a general sense that a search engine is "best" or personal preferences in search style, and in another 7.1% of sessions, participants indicate no specific reason (*No Reason*), suggesting automatic use. In 5% of the sessions, participants use SEs complementarily with a GenAI chatbot (*Complementary*). Finally, *Human Perspectives* (4.1%) are cited by participants who prefer SEs to access content created by real people.

Our analysis of 247 AI sessions reveals distinct patterns in participants' reasons for preferring GenAI chatbots over SEs, see Figure 2. For GenAI chatbots, participants most frequently cite synthesized or direct answers (*Content: Synthesized/Direct*, 22.3%), valuing GenAI chatbots' ability to directly summarize multiple sources into one response. *Efficiency* (19.4%) is similarly important, GenAI chatbots helping to quickly access and process substantial information, often under time pressure. *Functionality* (19.4%) captures preferences for capabilities not available in SEs, such as content creation (text, poems, images, code), or specific features like voice input. *Personalization* (17%) reflects valuing responses tailored to individual contexts and needs. Content quality (*Content: Other*, 13%) is another motivation, with participants describing AI responses as clearer, better, or more adequate, while *Content: Detailed* (9.3%) highlights an expectation for more comprehensive answers. The conversational nature of GenAI chatbots (*Conversational*, 8.1%) is appreciated for enabling iterative, interactive engagement. Skepticism toward SEs (*SE Skepticism*, 6.9%) stems from concerns about the accuracy and reliability of SE information, but also prior negative experiences due to sponsored SE results. *Other* reasons (6.5%) are generally vague or convenience-based, and *Complementary* (6.1%) indicates situations where AI is used alongside SEs. *Accessibility* (4%) reflects the immediate availability of GenAI chatbots in participants' workflow. *Technical Curiosity* (3.6%) captures exploratory use of AI, while *Humanoid Elements* (3.2%) indicates human-like attributions to GenAI chatbots, such as thinking or having a brain. Finally, *Familiarity/Habit* (0.8%) is rarely mentioned, suggesting AI use is more deliberate than automatic.

Overall, participants favor SEs for reliable, verifiable information and routine use, external source access, and variety. In contrast, GenAI chatbots are preferred for synthesized, ready-to-use, and personalized outputs, reducing the need for users to independently gather and integrate information.

4.1.2 Similar Categories, Different Meanings. A closer analysis of categories shows that several preference concepts are shared across tools but take on different meanings. *Efficiency* is valued in both contexts, but for SEs it refers to quick access to simple information, whereas for GenAI chatbots it reflects avoiding the need to independently collect and combine information from multiple sources. *Functionality* also diverges: SEs are valued for predictable, task-specific features such as navigation, media search, or shopping, while GenAI chatbots are valued for generative and interactive capabilities. Information quality is likewise interpreted differently. SE preferences emphasize relevance, accuracy, and verifiability, whereas GenAI chatbot preferences reflect a broader notion of "better" content, often without specifying the criteria underlying this assessment. *Accessibility* is also tool-specific: SEs are valued for general and situational convenience and due to technical defaults, while GenAI chatbots are valued for situational accessibility. *Skepticism* further differs across tools, with SEs evoking concerns about reliability, relevance, and commercialization, and GenAI chatbots eliciting skepticism mainly regarding reliability and relevance. Across categories, a broader pattern emerges in how users manage information across tools. *Source Variety* and *Short/Simple* in SEs reflect independent, user-controlled information gathering for straightforward tasks, whereas *Synthesized/Direct* and *Content: Detailed* in GenAI chatbots reflect system-mediated aggregation that delivers integrated information.

4.1.3 Tool-Specific Categories. Several categories appear (quasi-)exclusively for one tool type, highlighting distinct and largely non-overlapping value propositions.

For SE preferences, four categories are identified: (1) *External Access* (10.8%), capturing participants' use of search primarily as a navigation tool to reach known sources or specific websites rather than for direct answers; (2) *Human Perspectives* (4.1%), reflecting a preference for human-created content over algorithmic responses; (3) *No Reason* (7.1%), with users not being able to articulate a particular reason, suggesting unconscious usage; (4) *Familiarity/Habit* (13.3%,) emerging as a major preference category for SEs, while nearly absent for GenAI chatbots (0.8%). Taken together, *Familiarity/Habit* and *No Reason* indicate strong routine use, *External Access* highlights navigational gateway role, and *Human Perspectives* reflects the persistent need for human-authored content.

For GenAI chatbot preferences, the following four categories are tool-specific: (1) *Personalization* (17.0%), reflecting preference for contextually tailored responses; (2) *Conversational* (8.1%), highlighting interactive dialogue capabilities; (3) *Humanoid Elements* (3.2%), capturing anthropomorphized qualities and human-like interaction; (4) *Technical Curiosity* (3.6%), reflecting exploration of new technological capabilities. Together, these categories emphasize the interactive and conversational character of GenAI chatbots, with *Personalization*, *Conversational*, and *Humanoid Elements* describing typical user engagement, while *Technical Curiosity* shows that for some users, GenAI chatbots are still novel.

4.1.4 *Clustering of Preferences.*

Participants often provided multiple reasons per response, leading to co-occurrences. We used Louvain community detection to produce preference networks for SEs (Figure 3) and GenAI chatbots (Figure 4).

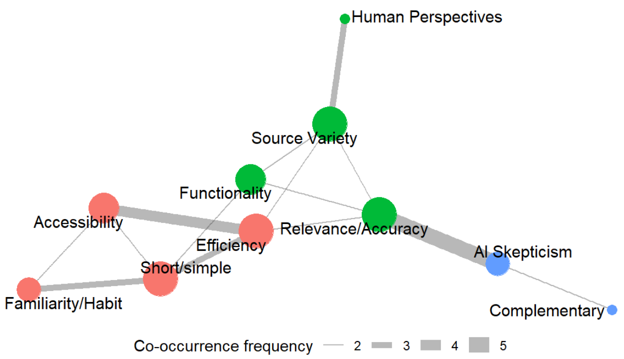

Figure 3: Network of preference clusters of SE preferences

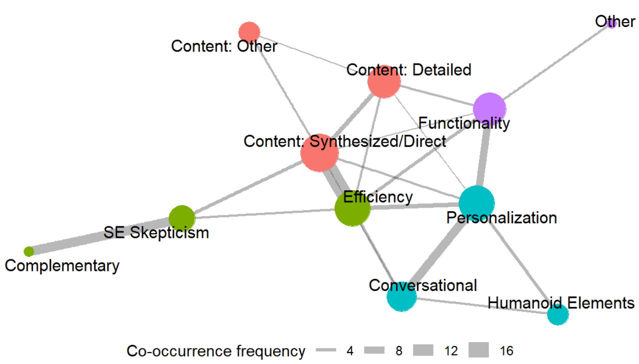

Figure 4: Network of preference clusters of AI preferences

SE preferences form a three-cluster structure:

1. **Content Characteristics:** Reasons centering on the nature and trustworthiness of the information retrieved, including *Relevance/Accuracy* (degrees $d = 4$), *Source Variety* ($d = 4$), *Human Perspectives* ($d = 1$), and *Functionality* ($d = 3$).
2. **Usability Factors:** Practical, procedural, and habitual reasons such as *Efficiency* ($d = 4$), *Familiarity/Habit* ($d = 2$), *Accessibility* ($d = 3$), and *Short/Simple* ($d = 4$).
3. **Tool Attitudes:** Broader, less specific motivations, including *Skepticism* ($d = 2$), *Complementary* ($d = 1$).

Cross-cluster analysis shows that Content Characteristics serves as the central hub, bridging the two other clusters. Specifically, the strongest connection is to Tool Attitudes through *Relevance/Accuracy* ↔ *AI Skepticism* (5 co-occurrences), suggesting that participants emphasizing accuracy also tend to be critical of GenAI chatbots. Content Characteristics also links to Usability Factors through multiple reason pairs, such as *Relevance/Accuracy* ↔ *Efficiency* (2), *Source Variety* ↔ *Efficiency* (2), and *Functionality* ↔ *Short/Simple* (2), highlighting how preferences for SEs combine trust in content with practical usability.

On the other hand, GenAI chatbot preferences form a more distributed four-cluster network:

1. **Usability Factors** : Overall usefulness and appeal of GenAI chatbots, including *Efficiency* ($d = 6$), *Complementary* ($d = 1$), and *SE Skepticism* ($d = 3$).
2. **Conversational Style**: Reasons emphasizing the conversational interaction style, including *Personalization* ($d = 6$), *Conversational* ($d = 4$), *Humanoid Elements* ($d = 2$).
3. **Core Functionality**: Specific capabilities often unique to GenAI chatbots, including *Functionality* ($d = 5$) and *Other* ($d = 1$).
4. **Content Characteristics**: Reasons focusing on the type of content that is generated, concerning its format, quality, and depth of information, including *Content: Synthesized/Direct* ($d = 7$), *Content: Detailed* ($d = 5$), and *Content: Other* ($d = 2$).

Preference reasons in the AI network are more interconnected across clusters. Two co-dominant hubs emerge: Usability Factors (36 connections) and Content Characteristics (35), with the strongest link between them via *Efficiency* ↔ *Content: Synthesized/Direct* (16 co-occurrences), highlighting participants' emphasis on AI efficiency for quickly receiving integrated information. Additional links, such as *Efficiency* ↔ *Content: Detailed* (3) and *SE Skepticism* ↔ *Content: Synthesized/Direct* (4), reinforce the value of rapid access to detailed, synthesized content. Conversational Style (26 connections) acts as an intermediary hub, bridging clusters through *Personalization* and *Conversational*. Its strongest external link is with Core Functionality via *Functionality* ↔ *Personalization* (8), showing participants perceive AI's core capabilities as closely tied to tailoring outputs to individual needs. It also connects to Usability Factors through *Efficiency* ↔ *Personalization* (5) and *Efficiency* ↔ *Conversational* (4), thus efficient AI use depends on responsive, personalized interaction. Links to Content Characteristics, including *Content: Synthesized/Direct* ↔ *Personalization* (3), *Content: Detailed* ↔ *Personalization* (2), and *Content: Synthesized/Direct* ↔ *Conversational* (2), emphasize that users value content for both comprehensiveness and contextual personalization. Core Functionality, though fewer in total connections (15), bridges all clusters through *Functionality*. Key links include *Functionality* ↔ *Personalization* (8), *Functionality* ↔ *Efficiency* (4), *Functionality* ↔ *Content: Detailed* (3), and *Functionality* ↔ *Content: Synthesized/Direct* (2).

4.1.5 *Tool Choice Philosophies.*

The preference analysis reveals distinct tool approaches: SEs support a 'Browse and Verify' approach, where users maintain control by reviewing and cross-checking multiple sources and prioritizing reliability and accuracy, whereas GenAI chatbots support a 'Delegate and Consume' approach, in which users rely on AI functionalities to synthesize and manage information with reduced user effort.

4.2 Topics, Triggers, Tasks, and Outcomes

After analyzing user preferences guiding tool choice, we next examine what users actually do with each tool, including topics, usage trigger situations, task types, and their outcomes. Across all dimensions, standardized residuals from χ^2 tests highlight where SEs and

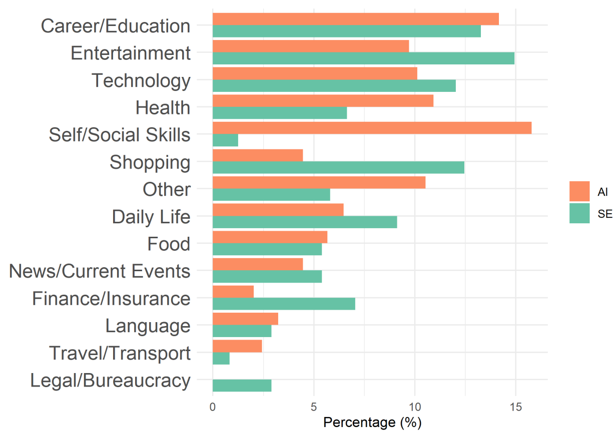

Figure 5: Distribution of topics for AI and SE sessions

GenAI chatbots differ significantly, absolute values >2 indicating significance.

4.2.1 Topics. Figure 5 shows the distribution of topics across both SE and AI sessions. SE sessions are dominated by Entertainment (14.9%; e.g. *"drake new album"*), Career/Education (13.3%; e.g. *"best tech skills in the workplace"*), and Shopping (12.4%; e.g. *"Samsung S25 Ultra case"*). AI sessions show somewhat different priorities: Self/Social Skills (15.8%; e.g. *"How long should I hold out for a response [after a date]"*), Career/Education (14.2%; e.g. *"how can I improve my communication skills?"*), and Health (10.9%; e.g. *"what vitamins should I take for bone health"*). Overall, the distributions of topics differ significantly between SE and AI sessions ($\chi^2 = 65.606$, $df = 13$, $p < 0.001$). Standardized residuals reveal that SEs are significantly more often used for Shopping ($r = 3.18$), Finance/Insurance ($r = 2.68$; e.g. *"city utilities payment"*), and Legal/Bureaucracy ($r = 2.70$; e.g. *"DMV near me"*), reflecting transactional or navigational topics. In contrast, AI sessions are more likely to involve Self/Social Skills ($r = 5.73$), which typically focus on interpersonal advice in often sensitive or uncertain social situations.

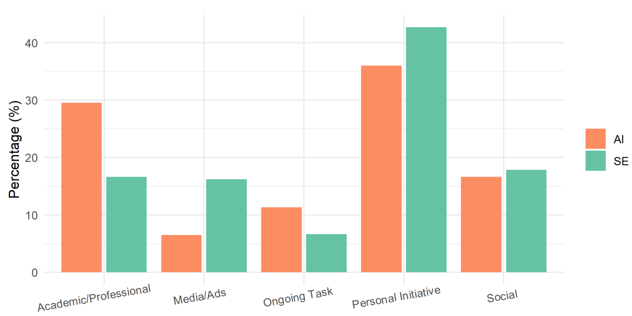

Figure 6: Distribution of triggers for AI and SE sessions

4.2.2 Triggers. We identify five types of trigger situations that initiate tool use: academic/professional contexts (study- or work-related needs), media and advertisements (encountering content in the news, ads, or email), social triggers (arising from a social interaction), ongoing tasks (while working on something not explicitly tied to study or work), and personal initiative (often stemming from curiosity or no specific triggers being mentioned). Figure 6 shows the distribution of these triggers across SE and AI sessions, which differ significantly overall ($\chi^2 = 23.526$, $df = 4$, $p < 0.001$).

In both modalities, sessions are primarily driven by personal initiative (SE: 42.7%, AI: 36.0%). Social triggers and ongoing tasks occur at similar rates across modalities.

Standardized residuals highlight the main differences. AI sessions are significantly more often triggered by academic/professional needs ($r = 3.39$; 29.6% vs 16.6%), consistent with guidance- or task-oriented use. In contrast, SE sessions are more frequently prompted by media or advertisements ($r = 3.39$; 16.2% vs 6.5%), reflecting reactive, stimulus-driven behavior.

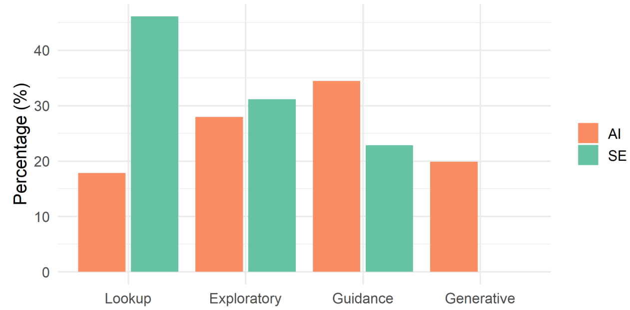

Figure 7: Distribution of tasks for AI and SE sessions

4.2.3 Tasks. The task dimension captures users' information intent, describing how they approach finding, exploring, or producing information during a session. We identify four task types:

1. *Lookup:* Quick sessions to find specific facts, known websites, or straightforward answers. Focused on one specific item with a clear goal.
2. *Exploratory:* Open-ended sessions aimed at gaining a broader understanding of a topic, often involving multiple queries or perspectives. The goal is not definite.
3. *Guidance:* Sessions to obtain advice, step-by-step instructions, or practical solutions to a problem or task. Solution-oriented and pragmatic.
4. *Generative:* Delegating a creative or cognitive task to the tool to produce content, such as the generation of text, code, or images.

Task distributions differ significantly between SEs and GenAI chatbots ($\chi^2 = 84.579$, $df = 3$, $p < 0.001$; the overall distribution across task types is shown in Figure 7). Standardized residuals show that SE sessions are dominated by lookup tasks ($r = 6.70$; 46.1%), nearly three times more frequent in SE than in AI sessions (17.8%), emphasizing the role of SEs for rapid fact-finding. AI sessions show higher rates of guidance tasks ($r = 2.83$; 34.3% vs 22.8%) and are the only modality with generative tasks ($r = 7.29$; 19.8% vs 0%). Exploratory tasks are similarly prevalent across both tools ($r = 0.77$, not significant; 31% vs 27.9%).

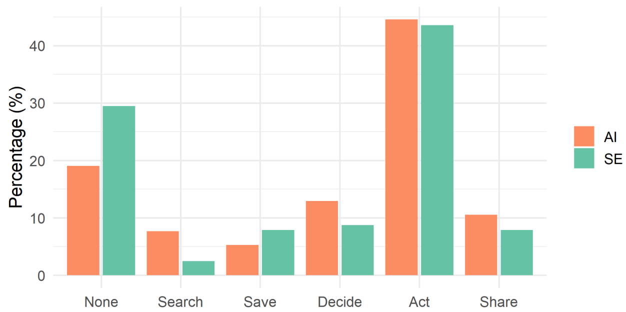

Figure 8: Distribution of outcomes for AI and SE sessions

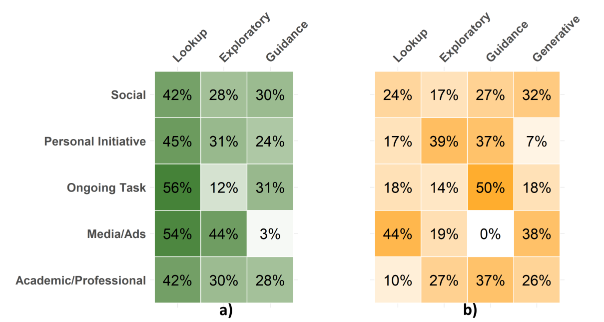

Figure 9: Task probability by trigger in a) SE sessions b) AI sessions

4.2.4 Outcomes. The outcome dimension captures what users report doing with the information retrieved after a session, focusing on concrete downstream actions rather than internal processes such as learning or reflection. Outcomes are ordered by increasing level of observable impact, ranging from no immediate action to sharing information with others. Specifically, outcomes include: *None* (no reported action beyond understanding or reflection), *Search* (conducting further searches to verify, compare, or elaborate on the information), *Save* (bookmarking, copying, or otherwise storing information for later use), *Decide* (reaching a decision or conclusion), *Act* (directly applying the information, such as following instructions or completing a task), and *Share* (passing the information on to others).

Outcome distributions differ significantly between SE and AI sessions ($\chi^2 = 16.183$, $df = 5$, $p < 0.001$). The overall distribution of outcomes across tools is shown in Figure 8. In both modalities, sessions most commonly result in direct action (Act; SE: 43.6%, AI: 44.5%). However, SE sessions more often end without any downstream action ($r = 2.69$; 29.5% vs. 19.0%), suggesting that SEs more frequently satisfy immediate information needs. In contrast, AI sessions are more likely to trigger additional searching ($r = 2.61$; 7.7% vs. 2.5%), indicating that AI tools more often stimulate follow-up inquiry or verification.

4.3 Session Flows

This section examines associations between triggers, tasks, and outcomes at the session level for SE and AI interactions.

4.3.1 Trigger-Task. In **SE sessions**, all trigger types most frequently lead to Lookup tasks (Figure 9). The overall association between triggers and tasks is not statistically significant ($\chi^2 = 14.11$, $p = 0.079$, Cramér's V = 0.171), thus triggers do not systematically influence task choice. However, post-hoc pairwise comparisons reveal a notable exception: Media/Ads triggers are less likely than all other trigger types to result in Guidance tasks (all $p < 0.05$).

In **AI sessions**, Ongoing Task triggers are most likely to lead to Guidance tasks (50%), Social triggers show a balanced distribution across all task types, and Personal Initiative sometimes triggers Generative tasks (6.8%). Media/Ads never triggers Guidance tasks and instead trigger Lookup (43.8%) and Generative (37.5%; see Figure 9). The overall association between trigger type and task choice is statistically significant ($\chi^2 = 41.16$, $p < 0.001$, Cramér's V = 0.236). Post-hoc pairwise comparisons show that the Media/Ads trigger is significantly more likely to lead to Lookup tasks than Academic/Professional ($p = 0.016$), but Media/Ads is less likely than Academic/Professional ($p = 0.016$), Ongoing Task (both $p = 0.010$), and Personal Initiative ($p = 0.016$) to trigger Guidance tasks. Personal Initiative is significantly less likely than Academic/Professional ($p = 0.012$), Media/Ads (both $p = 0.016$), and Social triggers ($p = 0.010$) to trigger Generative tasks.

4.3.2 Task-Outcome. In **SE sessions**, task type strongly shapes outcomes (Figure 10). Guidance tasks are highly action-oriented, with 69.1% resulting in direct action. Exploratory tasks most often end without follow-up (40.0%), while Lookup tasks fall in between, frequently leading to action (44.1%) but also often without further steps (33.3%). The association between tasks and outcomes is statistically significant ($\chi^2 = 42.49$, $p < 0.001$, Cramér's V = 0.297). Post-hoc comparisons confirm that both Lookup and Exploratory tasks are significantly more likely than Guidance tasks to result in no action ($p \leq 0.001$), Lookup tasks are more likely than Exploratory tasks to lead to direct action ($p < 0.019$) while Guidance tasks are significantly more likely than both Lookup ($p = 0.013$) and Exploratory ($p < 0.001$) to result in direct action.

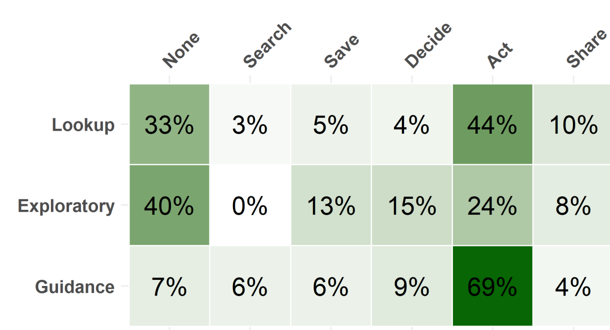

Figure 10: Outcome probability by task in SE sessions

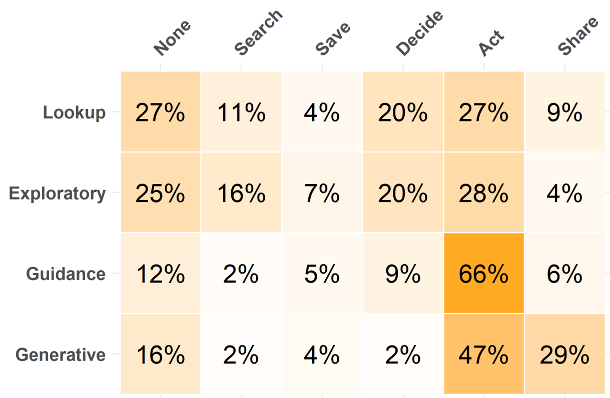

Figure 11: Outcome probability by task in AI sessions

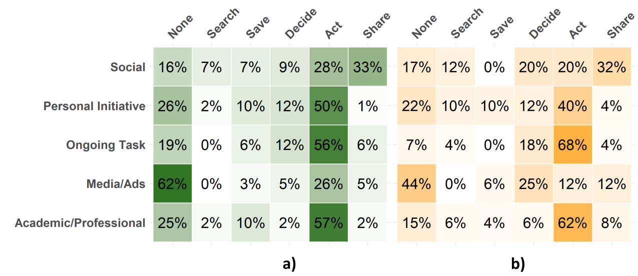

Figure 12: Outcome probability by trigger in a) SE sessions b) AI sessions

In **AI sessions**, task type also shapes outcomes (Figure 11). Guidance tasks most frequently result in direct action (65.9%), while Generative tasks lead to action in 46.9% of cases and show the highest sharing probability (28.6%). Lookup and Exploratory tasks display more distributed outcomes with very similar patterns. The association between tasks and outcomes is statistically significant ($\chi^2 = 63.83$, $p < 0.001$, Cramér's V = 0.294). Post-hoc comparisons indicate that Guidance tasks are significantly more likely than Lookup and Exploratory tasks to result in action ($p < 0.001$), Exploratory tasks are more likely than Guidance tasks to lead to further searching ($p = 0.022$), Lookup and Exploratory tasks are more likely than Generative tasks to result in decision-making ($p \leq 0.029$), and Generative tasks are more likely than both Exploratory and Guidance tasks to result in sharing ($p \leq 0.005$).

4.3.3 Trigger-Outcome. In **SE sessions**, downstream outcomes systematically vary depending on the trigger that initiated the session (Figure 12). Media/Ads triggers are most likely to result in no follow-up action (61.5%), whereas Academic/Professional (57.5%), Personal Initiative (49.5%), and Ongoing Task (56.2%) triggers frequently lead to direct action. Social triggers show a distinct sharing pattern, with 32.6% of sessions resulting in information being shared. The association between triggers and outcomes is statistically significant ($\chi^2 = 78.06$, $p < 0.001$, Cramér's V = 0.285). Post-hoc comparisons indicate that Media/Ads triggers are significantly more likely than Academic/Professional ($p = 0.017$), Personal Initiative (0.003), and Social triggers ($p < 0.001$) to result in no follow-up action, while Social triggers are significantly more likely than all other trigger types to result in sharing ($p \leq 0.019$).

In **AI sessions**, outcome patterns also vary systematically by trigger type (Figure 12). Sessions triggered by Ongoing Tasks and Academic/Professional contexts are the most action-oriented, with 67.9% and 61.6% respectively, resulting in direct action. In contrast, Media/Ads triggers show a markedly more passive pattern, with low rates of action (12.5%) and a high proportion of sessions ending without further actions (43.8%). Social triggers display a characteristic sharing pattern, with 31.7% resulting in information being shared. These differences are supported by a statistically significant association between triggers and outcomes ($\chi^2 = 67.46$, $p < 0.001$, Cramér's V = 0.261). Post-hoc comparisons confirm that Academic/Professional and Ongoing Task triggers are significantly more likely than Media/Ads and Social triggers to result in direct action ($p \leq 0.006$), while Social triggers are significantly more likely than all other trigger types to lead to sharing information ($p \leq 0.043$). Media/Ads triggers are also more likely than Ongoing Task triggers to result in no subsequent action ($p = 0.050$).

5 Discussion

Our comparative analysis reveals that, despite the growing functional overlap between SEs and GenAI chatbots, users do not treat them as interchangeable. They maintain distinct mental models, expectations, and behavioral patterns for each tool. By grounding our study in participants' reflections on their actual interaction logs, we capture authentic behaviors and reasoning, moving beyond hypothetical scenarios or isolated log analyses to reveal real-world information-seeking patterns.

User Philosophies. Participants articulate two distinct philosophies when approaching each system. SEs are preferred for accuracy, reliability, and verifiability, supporting independent evaluation and comparison across sources, a 'Browse and Verify' approach. Conversational GenAI chatbots, by contrast, are valued for synthesized, personalized, and generative outputs, enabling efficient handling of complex information, a 'Delegate and Consume' approach. These preferences reflect users' expectations about control, effort, and the trustworthiness of each tool.

Actual Usage Patterns. SEs are primarily used for quick fact-finding on transactional or navigational topics such as Shopping, Finance, and Legal/Bureaucracy, often triggered by media exposure or personal initiative. Many searches end without further action, reinforcing SEs' role as rapid IR tools. This aligns with prior work showing that many needs are resolved directly on SERPs [17]. In contrast, GenAI chatbots are frequently used for guidance-seeking and generative tasks, particularly in domains such as health, social skills, and academic/professional contexts, leading to real-world actions, where prior research has highlighted concerns about over-reliance and misinformation [52, 61].

Philosophy-Behavior Misalignment. Placing these findings side by side hints at an asymmetric pattern between users' articulated philosophies and their enacted practices. SE use aligns closely with the 'Browse and Verify' philosophy: users seek accurate, verifiable information and retain control over evaluation, consistent with how they describe preferring SEs. GenAI chatbot use tells a more complex story. While users articulate a 'Delegate and Consume' approach centered on efficiency and synthesis, in practice, they frequently turn to GenAI chatbots for sensitive topics such as health concerns, relationship advice, and personal dilemmas, domains where reliability is arguably most critical. Users also often act on the guidance received, potentially carrying real-world consequences if misinformation or overreliance occurs.

5.1 Implications

Our findings suggest several implications for the evaluation and benchmarking of IR systems. Despite the increasing convergence of SEs and GenAI chatbots, users appear to maintain distinct mental models and behavioral patterns for each. We therefore propose tool-specific evaluation criteria that reflect current user expectations, while noting that as these systems continue to converge, criteria may increasingly need to travel across tools.

SE benchmarking suites should include navigational and transactional query types, and evaluation metrics should reward source diversity, verifiability, and retrieval speed over elaborate synthesis. As generative features are integrated into SEs, evaluation frameworks should explicitly test whether hybrid systems preserve search-specific strengths: clear source attribution, result diversity, and support for user-driven comparison and verification.

For AI-powered retrieval systems, benchmarking tasks should be grounded in the consequential domains where users actually deploy these tools, such as health, relationship advice, and personal decision-making, rather than relying solely on factoid or knowledge-intensive tasks that dominate current benchmarks. Concretely, evaluation metrics should go beyond output quality and fluency to capture credibility mechanisms: source attribution, confidence signaling, and explainability. Safety and reliability audits should specifically target high-stakes domains where users seek actionable guidance, given the real-world consequences of misinformation or overreliance in these contexts.

5.2 Limitations and Future Work

Several limitations should be considered when interpreting our findings. First, our Prolific sample is not fully representative of the general population of internet users, as participants tend to have higher digital literacy and more extensive online experience [48]. This is further reinforced by our study design, which required participants to access and reflect on their own interaction histories, thereby implicitly excluding individuals with limited ability to navigate platform data or account settings. This may limit generalizability to less digitally literate populations, who may interact with GenAI chatbots differently, including potentially lower awareness of system limitations. Second, our reliance on self-reported data introduces well-known biases, including post-hoc rationalization [58]. Although our design explicitly anchored participants' reflections in their past search histories to reduce recall bias, this does not fully eliminate reconstruction effects in self-reports. In addition, uneven recall across SE and AI sessions, for instance, due to differences in usage frequency or recency effects, may introduce recall bias in reported behaviors. Third, focusing on individual sessions provides detailed insights into specific interactions but does not capture multi-tool strategies or cross-session behaviors. Finally, the rapidly evolving information ecosystem limits the temporal generalizability of our findings. During data collection (February 2025), features such as AI Overviews in search engines and search-features within GenAI chatbots were only beginning to be integrated. Since then, further integration, such as Google's AI Mode, has continued to blur distinctions between search engines and generative GenAI chatbots. While our study captures a meaningful snapshot of a period in which these tools were still relatively distinguishable, user practices and platform affordances are likely to have evolved.

Future research could extend this work in several directions. First, it could examine how less digitally literate users engage with search engines compared to GenAI chatbots, particularly how awareness of system limitations and critical evaluation of outputs may differ across user populations. Second, combining trace or log-based data (e.g., through data donation) with participant reflections could provide a more comprehensive view of actual behavior alongside perceived behavior. Third, studying complete information-seeking episodes across multiple tools would enable a better understanding of cross-tool and multi-session strategies in realistic workflows. Finally, future work could investigate how evolving system capabilities shape users' mental models, trust, and information evaluation practices, including source verification and fact-checking behaviors.

6 Conclusion

This comparative study of search engines and GenAI chatbots examines both users' stated preferences and their actual information-seeking behavior. Tool choice preferences reveal distinct philosophies: SEs embody a 'Browse and Verify' approach, emphasizing accuracy, source variety, and independent evaluation, while GenAI chatbots reflect a 'Delegate and Consume' approach, prioritizing efficiency, synthesis, and personalization. An analysis of actual usage sessions shows that SE behavior aligns closely with this philosophy, supporting rapid fact-finding on transactional and navigational topics. GenAI chatbot use, however, hints at a more complex picture: while valued for efficiency and synthesis, users also frequently turn to GenAI chatbots for actionable guidance in sensitive domains such as health and social skills, suggesting a potential gap between stated preferences and enacted behavior. These findings point to the importance of IR system development that preserves reliability and user control in SEs, while integrating transparency and verification mechanisms in AI-driven retrieval.

Acknowledgments

This work was supported by funding from the Swiss National Science Foundation (Grant number 215354).

References

[1] Beatriz Botero Arcila. 2023. Is it a platform? is it a search engine? it's chatgpt! the european liability regime for large language models. *J. Free Speech L.*, 3, 455.

[2] Emily M Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the dangers of stochastic parrots: can language models be too big? In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*, 610–623.

[3] Markus Bink, Steven Zimmerman, and David Elsweiler. 2022. Featured snippets and their influence on users' credibility judgements. In *Proceedings of the 2022 Conference on Human Information Interaction and Retrieval*, 113–122.

[4] Vincent D Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. 2008. Fast unfolding of communities in large networks. *Journal of statistical mechanics: theory and experiment*, 2008, 10, P10008.

[5] Niall Bolger, Angelina Davis, and Eshkol Rafaeli. 2003. Diary methods: capturing life as it is lived. *Annual review of psychology*, 54, 1, 579–616.

[6] Kenrick Cai, Krystal Hu, and Anna Tong. 2025. Exclusive: openai to release web browser in challenge to google chrome. (July 2025). https://www.reuters.com/business/media-telecom/openai-release-web-browser-challenge-google-chrome-2025-07-09/.

[7] Noel Carroll. 2014. In search we trust: exploring how search engines are shaping society. *International Journal of Knowledge Society Research (IJKSR)*, 5, 1, 12–27.

[8] Aaron Chatterji, Thomas Cunningham, David J Deming, Zoe Hitzig, Christopher Ong, Carl Yan Shan, and Kevin Wadman. 2025. How people use chatgpt. Tech. rep. National Bureau of Economic Research.

[9] Danqi Chen, Weizhu Chen, Haixun Wang, Zheng Chen, and Qiang Yang. 2012. Beyond ten blue links: enabling user click modeling in federated web search. In *Proceedings of the fifth ACM international conference on Web search and data mining*, 463–472.

[10] Jia Chen, Jiaxin Mao, Yiqun Liu, Fan Zhang, Min Zhang, and Shaoping Ma. 2021. Towards a better understanding of query reformulation behavior in web search. In *Proceedings of the web conference 2021*, 743–755.

[11] Gábor Csárdi and Tamás Nepusz. 2006. The igraph software package for complex network research. *InterJournal*, Complex Systems, 1695. https://igraph.org.

[12] Robert Epstein and Ronald E Robertson. 2015. The search engine manipulation effect (seme) and its possible impact on the outcomes of elections. *Proceedings of the national academy of sciences*, 112, 33, E4512–E4521.

[13] European Commission. 2018. Antitrust: commission fines google €4.34 billion for illegal practices regarding android mobile devices to strengthen dominance of google search. Press release IP/18/4581. (July 2018). https://ec.europa.eu/commission/presscorner/detail/en/ip_18_4581.

[14] Erik Fubel, Niclas Michael Groll, Patrick Gundlach, Qiwei Han, and Maximilian Kaiser. 2023. Beyond rankings: exploring the impact of serp features on organic click-through rates. *arXiv preprint arXiv:2306.01785*.

[15] Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Houda Bouamor, Juan Pino, and Kalika Bali, (Eds.) Association for Computational Linguistics, Singapore, (Dec. 2023), 6465–6488. doi:10.18653/v1/2023.emnlp-main.398.

[16] Tarleton Gillespie. 2014. The relevance of algorithms. *Media technologies: Essays on communication, materiality, and society*, 167, 2014, 167.

[17] Jeffrey Gleason, Desheng Hu, Ronald E Robertson, and Christo Wilson. 2023. Google the gatekeeper: how search components affect clicks and attention. In *Proceedings of the International AAAI Conference on Web and Social Media*. Vol. 17, 245–256.

[18] Eszter Hargittai and Yuli Patrick Hsieh. 2012. Succinct survey measures of web-use skills. *Social Science Computer Review*, 30, (Feb. 2012), 95–107, 1, (Feb. 2012). doi:10.1177/0894439310397146.

[19] Desheng Hu, Jeffrey Gleason, Muhammad Abu Bakar Aziz, Alice Koeninger, Nikolas Guggenberger, Ronald E Robertson, and Christo Wilson. 2024. Market or markets? investigating google search's market shares under vertical segmentation. In *Proceedings of the International AAAI Conference on Web and Social Media*. Vol. 18, 637–650.

[20] Krystal Hu. 2023. Chatgpt sets record for fastest-growing user base. ChatGPT reached 100 million users in just two months after its November 2022 release, becoming the fastest-growing consumer app in history. (Feb. 2023).

[21] Martin Huschens, Martin Briesch, Dominik Sobania, and Franz Rothlauf. 2023. Do you trust chatgpt?–perceived credibility of human and ai-generated content. *arXiv preprint arXiv:2309.02524*.

[22] Bernard J. Jansen, Amanda Spink, Chris Blakely, and Sherry Koshman. 2007. Defining a session on web search engines. *Journal of the American Society for Information Science and Technology*, 58, (Apr. 2007), 862–871, 6, (Apr. 2007). doi:10.1002/asi.20564.

[23] Ehsan Kamalloo, Aref Jafari, Xinyu Zhang, Nandan Thakur, and Jimmy Lin. 2023. Hagrid: a human-llm collaborative dataset for generative information-seeking with attribution. *arXiv preprint arXiv:2307.16883*.

[24] Martin Kroh. 2007. Measuring left-right political orientation: the choice of response format. *The Public Opinion Quarterly*, 71, 2, 204–220. Retrieved May 12, 2025 from http://www.jstor.org/stable/4500371.

[25] Weronika Łajewska, Damiano Spina, Johanne Trippas, and Krisztian Balog. 2024. Explainability for transparent conversational information-seeking. In *Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval*, 1040–1050.

[26] Elsa Lichtenegger, Aleksandra Urman, and Aniko Hannak. 2026. A new taxonomy of web search: a user-centered framework for search intent in the ai era. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems* (CHI '26) Article 692. Association for Computing Machinery, 16 pages. ISBN: 9798400722783. doi:10.1145/3772318.3791050.

[27] Elsa Lichtenegger, Aleksandra Urman, and Aniko Hannak. 2026. Beyond the query: a survey design for eliciting underlying motivations in web search. In *Proceedings of the 2026 Conference on Human Information Interaction and Retrieval* (CHIIR '26). Association for Computing Machinery, 464–468. ISBN: 9798400724145. doi:10.1145/3786304.3787889.

[28] Lin Liu, Jiajun Meng, and Yongliang Yang. 2024. Llm technologies and information search. *Journal of Economy and Technology*, 2, 269–277.

[29] Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2022. When not to trust language models: investigating effectiveness of parametric and non-parametric memories. *arXiv preprint arXiv:2212.10511*.

[30] Gary Marchionini. 2006. Exploratory search: from finding to understanding. *Communications of the ACM*, 49, 4, 41–46.

[31] David McCabe and Michael J. Kramer. 2023. Google and apple face antitrust scrutiny over search spotlight. Accessed: 2025-08-14. (Oct. 2023). https://www.nytimes.com/2023/10/26/technology/google-apple-search-spotlight.html.

[32] Shahan Ali Memon and Jevin D West. 2024. Search engines post-chatgpt: how generative artificial intelligence could make search less reliable. *arXiv preprint arXiv:2402.11707*.

[33] Rachel Metz. 2025. Openai lets users go shopping with chatgpt, challenging google. (Apr. 2025). https://www.bloomberg.com/news/articles/2025-04-28/openai-lets-users-go-shopping-with-chatgpt?embedded-checkout=true.

[34] Luise Metzger, Linda Miller, Martin Baumann, and Johannes Kraus. 2024. Empowering calibrated (dis-) trust in conversational agents: a user study on the persuasive power of limitation disclaimers vs. authoritative style. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, 1–19.

[35] Eliza Mitova, Sina Blassnig, Edina Strikovic, Aleksandra Urman, Aniko Hannak, Claes de Vreese, and Frank Esser. 2025. Exploring public attitudes toward generative ai for news across four countries. *Journal of Quantitative Description: Digital Media*, 5.

[36] Safiya Umoja Noble. 2018. Algorithms of oppression: how search engines reinforce racism. In *Algorithms of oppression*. New York university press.

[37] Alamir Novin and Eric Meyers. 2017. Making sense of conflicting science information: exploring bias in the search engine result page. In *Proceedings of the 2017 conference on conference human information interaction and retrieval*, 175–184.

[38] Bruno Oliveira and Carla Teixeira Lopes. 2023. The evolution of web search user interfaces-an archaeological analysis of google search engine result pages. In *Proceedings of the 2023 conference on human information interaction and retrieval*, 55–68.

[39] Siru Ouyang et al. 2023. The shifted and the overlooked: a task-oriented investigation of user-gpt interactions. *arXiv preprint arXiv:2310.12418*.

[40] Bing Pan, Helene Hembrooke, Thorsten Joachims, Lori Lorigo, Geri Gay, and Laura Granka. 2007. In google we trust: users' decisions on rank, position, and relevance. *Journal of computer-mediated communication*, 12, 3, 801–823.

[41] Thomas Lin Pedersen. 2025. *ggraph: An Implementation of Grammar of Graphics for Graphs and Networks*. R package version 2.2.2.9000. https://ggraph.data-imaginist.com.

[42] Thomas Lin Pedersen. 2025. *tidygraph: A Tidy API for Graph Manipulation*. R package version 1.3.1.9000, https://github.com/thomasp85/tidygraph. https://tidygraph.data-imaginist.com.

[43] R Core Team. 2021. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/.

[44] Elizabeth Reid. 2024. Generative ai in search: let google do the searching for you. (May 2024). https://blog.google/products/search/generative-ai-google-search-may-2024/.

[45] Johannes Schneider. 2025. Mental model shifts in human-llm interactions. *Journal of Intelligent Information Systems*, 1–16.

[46] Phillip Schneider, Anum Afzal, Juraj Vladika, Daniel Braun, and Florian Matthes. 2023. Investigating conversational search behavior for domain exploration. In *European Conference on Information Retrieval*. Springer, 608–616.

[47] SEMrush. 2025. Most visited websites in the world, updated june 2025. Accessed: 2025-08-14. (2025). https://www.semrush.com/website/top/.

[48] Aaron Shaw and Eszter Hargittai. 2021. Do the online activities of amazon mechanical turk workers mirror those of the general population? a comparison of two survey samples. *International Journal of Communication*, 15, 16.

[49] SimilarWeb. 2025. Top websites ranking - most visited websites in the world. Accessed: 2025-08-14. (2025). https://www.similarweb.com/top-websites/.

[50] Cornelia Sindermann et al. 2021. Assessing the attitude towards artificial intelligence: introduction of a short measure in german, chinese, and english

language. *KI - Künstliche Intelligenz*, 35, (Mar. 2021), 109–118, 1, (Mar. 2021). doi:10.1007/s13218-020-00689-0.
[51] Andre Smith. 2025. What is perplexity ai: complete guide to the ai search engine 2025. Overview of Perplexity AI as a conversational, citation-rich search engine. (July 2025). https://screenapp.io/blog/what-is-perplexity-ai.
[52] Sofia Eleni Spatharioti, David Rothschild, Daniel G Goldstein, and Jake M Hofman. 2025. Effects of llm-based search on decision making: speed, accuracy, and overreliance. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 1–15.
[53] StatCounter. 2025. Ai chatbot market share united states of america. https://gs.statcounter.com/ai-chatbot-market-share/all/united-states-of-america. Accessed: 2026-04-27. (2025).
[54] StatCounter. 2025. Search engine market share united states of america. https://gs.statcounter.com/search-engine-market-share/all/united-states-of-america. Accessed: 2026-04-27. (2025).
[55] Robby Stein. 2025. Expanding ai overviews and introducing ai mode. (Mar. 2025). https://blog.google/products/search/ai-mode-search/.
[56] Siddharth Suri et al. 2024. The use of generative search engines for knowledge work and complex tasks. *arXiv preprint arXiv:2404.04268*.
[57] The Economic Times. 2024. Openai releases chatgpt search engine, taking on google. https://economictimes.indiatimes.com/tech/artificial-intelligence/openai-releases-chatgpt-search-engine-taking-on-google/articleshow/114830420.cms?from=mdr.
[58] Roger Tourangeau, Lance J Rips, and Kenneth Rasinski. 2000. The psychology of survey response.
[59] Vincent A Traag, Ludo Waltman, and Nees Jan Van Eck. 2019. From louvain to leiden: guaranteeing well-connected communities. *Scientific reports*, 9, 1, 1–12.
[60] Aleksandra Urman, Aniko Hannak, and Mykola Makhortykh. 2024. User attitudes to content moderation in web search. *Proceedings of the ACM on Human-Computer Interaction*, 8, (Apr. 2024), 1–27, CSCW1, (Apr. 2024). doi:10.1145/3637423.
[61] Aleksandra Urman and Mykola Makhortykh. 2025. The silence of the llms: cross-lingual analysis of guardrail-related political bias and false information prevalence in chatgpt, google bard (gemini), and bing chat. *Telematics and Informatics*, 96, 102211.
[62] Aleksandra Urman, Mykola Makhortykh, and Roberto Ulloa. 2022. The matter of chance: auditing web search results related to the 2020 us presidential primary elections across six search engines. *Social science computer review*, 40, 5, 1323–1339.
[63] Pranav Narayanan Venkit, Philippe Laban, Yilun Zhou, Yixin Mao, and Chien-Sheng Wu. 2024. Search engines in an ai era: the false promise of factual and verifiable source-cited responses. *arXiv preprint arXiv:2410.22349*.
[64] Laura Weidinger et al. 2022. Taxonomy of risks posed by language models. In *Proceedings of the 2022 ACM conference on fairness, accountability, and transparency*, 214–229.
[65] Ryen W White and Resa A Roth. 2009. *Exploratory search: Beyond the query-response paradigm*. Number 3. Morgan & Claypool Publishers.
[66] Hong Xie. 2002. Patterns between interactive intentions and information-seeking strategies. *Information processing and Management*, 38, 1, 55–77.
[67] Ruiyun Xu, Yue Feng, and Hailiang Chen. 2023. Chatgpt vs. google: a comparative study of search performance and user experience. *arXiv preprint arXiv:2307.01135*.
[68] Xi Ye, Ruoxi Sun, Sercan Arik, and Tomas Pfister. 2024. Effective large language model adaptation for improved grounding and citation generation. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 6237–6251.
[69] Jiawei Zhou, Yixuan Zhang, Qianni Luo, Andrea G Parker, and Munmun De Choudhury. 2023. Synthetic lies: understanding ai-generated misinformation and evaluating algorithmic and human solutions. In *Proceedings of the 2023 CHI conference on human factors in computing systems*, 1–20.